

# Detecting Clinical and Social Impacts of Substance Use in Social Media: A Comparative Study of Encoder-, LLM-, and SVM-Based NER Pipelines

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**Abstract**—Nonmedical opioid use is a persistent public-health issue across most developed countries, and the anonymity afforded by social media leads many affected individuals to share first-person testimonies that mention both *clinical impacts* (e.g., withdrawal, addiction, the need for medical assistance) and *social impacts* (e.g., legal trouble, family conflict, job loss, financial hardship). Automatically detecting these mentions in user-generated text is a meaningful first step toward early intervention and reintegration support. In this work, we investigate token-level Named Entity Recognition for clinical and social impacts of substance use in Reddit posts, in the context of the SMM4H-HeaRD 2026 Task 7 shared task, and compare four families of approaches: (i) a direct pipeline, in which a single encoder produces 5-class BIO tags, evaluated across several pretrained backbones, loss functions, and an optional CRF decoding head; (ii) a two-stage pipeline, in which a 3-class encoder detects impact spans and a Large Language Model subsequently classifies each span as Clinical, Social, or Neither, in both zero-shot and few-shot regimes; (iii) an audit pipeline, in which an LLM verifies or corrects, span by span, the predictions of a 5-class encoder (again zero- and few-shot); and (iv) a lightweight classical-ML variant, in which the LLM in pipelines (ii) and (iii) is replaced by an SVM head trained on encoder span embeddings. Our best configuration, a DeBERTa-v3 encoder with a CRF decoding head, attains 45.38% strict micro-F1 and 54.20% relaxed overlap F1 on the official Codabench test set. None of our configurations surpasses the shared-task baseline reported by Dey *et al.* [1], but the cross-pipeline ablation yields concrete evidence on where encoder-only models, LLM-based type classification, LLM-based auditing, and lightweight SVM classifiers each help or hurt.

**Index Terms**—Named Entity Recognition, substance use, opioid epidemic, social media, BIO tagging, Conditional Random Fields, Large Language Models, few-shot prompting, SMM4H-HeaRD.

## I. INTRODUCTION

**N**ONMEDICAL use of prescription and street opioids is a public-health problem in most developed countries, with a high burden among young and working-age adults. Epidemiological studies have characterised the affected demographics and their social drivers, but tools for the early identification, prevention, and remediation of opioid-related harms remain limited. Over the last years a new source of data has emerged: the personal accounts that users post on social media. The anonymity offered by platforms such as Reddit lowers the barrier to self-disclosure, so affected individuals

describe their substance-use experiences in detail and without fear of stigma. Social media therefore forms a rich corpus for training systems that flag users who have already suffered substance-use consequences — a first step toward timely intervention.

This work presents a comparative study of methods for extracting clinical and social impacts of nonmedical opioid use from first-person social-media posts. The study is conducted in the context of the SMM4H-HeaRD 2026 Task 7 shared task, which provides a corpus of Reddit posts annotated under a BIO tagging scheme with two entity types: *ClinicalImpacts* — physical or psychological consequences such as overdose, withdrawal, depression, or anxiety — and *SocialImpacts* — relational, occupational, or legal consequences such as job loss, legal charges, or financial hardship. Only *first-person, self-reported* impacts are annotated, which makes the task harder than generic biomedical NER: a system has to separate confessional posts (“*I withdrew really bad*”) from third-person reports (“*my friend overdosed*”), which must remain untagged. Performance is reported with two metrics: *strict micro-F1*, which requires exact agreement on both boundaries and label, and *relaxed overlap F1*, which credits any non-empty overlap with a gold span of the same type.

Within this setup, we organise our study around two research questions:

**RQ1.** Does decoupling *span detection* from *span typing*, by first locating impact spans with a 3-class encoder and then assigning their type with a downstream classifier, improve span-level F1 over a single 5-class encoder trained end-to-end?

**RQ2.** When span detection is already performed by a strong 5-class encoder, can a Large Language Model audit and correct its predictions, and is the gain (if any) larger than what a Support Vector Machine head can achieve on the same encoder span embeddings?

To answer these questions we evaluate four families of pipelines: (i) a **direct** pipeline, in which a fine-tuned encoder produces 5-class BIO predictions, with an optional Conditional Random Field decoding head; (ii) a **two-stage** pipeline, in which a 3-class encoder detects impact spans and an LLM labels each span as Clinical, Social, or Neither; (iii) an **audit** pipeline, in which an LLM verifies and, when needed, corrects the predictions of a 5-class encoder, span by span; and (iv) an **SVM-based variant**, in which the LLM in (ii) and (iii) is replaced by a Support Vector Machine head trained on

The full source code, configuration files, and trained-model checkpoints used to produce the experiments reported in this paper are publicly available at <https://github.com/vl4dio4n/SMM4H-HeaRD-2026-Task-7>.

encoder span embeddings. Each family is run across several encoder backbones, training losses, and both zero-shot and few-shot LLM regimes. None of our configurations surpasses the DeBERTa-large baseline reported by the shared-task organisers [1], but the cross-pipeline ablation addresses both research questions and shows where each design choice helps or hurts.

The remainder of the paper is organised as follows. Section II reviews related work and the building blocks of our pipelines. Section III describes the dataset and the design of each system. Section IV reports the experimental setup, Section V the results and a comparison across methods, and Section VI concludes with limitations and directions for future work.

## II. RELATED WORK

The work in this paper sits at the intersection of three strands of research: (i) domain-specific NER for substance-use text, (ii) pretrained transformer encoders fine-tuned for token classification, and (iii) instruction-tuned Large Language Models used as in-context classifiers. We review them in this order before describing how the field falls short.

### A. Substance-use NER and the SMM4H-HeaRD task

Public-health monitoring on social media has been studied for many years, but most work targets symptom or drug-name extraction rather than self-reported consequences. The dataset and the baseline used in this paper come from Dey *et al.* [1], who introduced the SMM4H-HeaRD 2026 Task 7 corpus of Reddit posts annotated with first-person clinical and social impacts of nonmedical opioid use, and who report a fine-tuned DeBERTa-large baseline at 0.61 relaxed F1. The same authors document a gap between expert annotators and current models: clinical professionals reach high agreement, while encoder-only systems struggle with implicit and figurative descriptions of impacts (e.g. “I lost everything”). Their dataset is small (around 1,700 annotated segments), imbalanced toward the `O` class, and biased toward English-speaking subreddits, which limits generalisation. Following Tjong Kim Sang and De Meulder [8], both the dataset and our evaluation adopt BIO tagging and span-level F1; we additionally distinguish *strict* F1 (exact span and label match) from *relaxed* F1 (any non-empty overlap with a gold span of the same type), the convention used by the shared task and implemented by `seqeval`.

### B. Pretrained transformer encoders

All four of our pipelines build on the encoder fine-tuning recipe introduced with BERT [2] and refined by RoBERTa [3], which removes the next-sentence prediction objective and trains for longer on more data. Our baseline encoder (Exp 01–02b) is `mental-roberta-base`, the RoBERTa checkpoint domain-adapted to mental-health Reddit text by Ji *et al.* alongside MentalBERT [5]; the same authors show that mental-health domain adaptation helps on Reddit-style text, which motivates this choice for an opioid-related corpus. Our second encoder is DeBERTa-v3 [4], which combines disentangled

attention with ELECTRA-style replaced-token-detection pre-training and outperforms RoBERTa on GLUE and Super-GLUE; this is the backbone for Experiments 03–05 and 09–10.

### C. Sequence labelling and CRF decoding

Conditional Random Fields [6] are the standard probabilistic model for joint sequence labelling: they assign a globally normalised score to a full label sequence and admit exact decoding via the Viterbi algorithm. Lample *et al.* [7] introduced the canonical recipe of pairing a neural encoder with a linear-chain CRF head to enforce valid BIO transitions (e.g. `I-Clinical` cannot follow `B-Social` or `O`). Our Exp 04 follows the same pattern, replacing the original BiLSTM with DeBERTa-v3, and serves as the strongest encoder-only system in our ablation.

### D. Training techniques on small, imbalanced data

Token-level NER on small and imbalanced datasets benefits from two training techniques that are now common in the literature. Focal loss [9], originally proposed for dense object detection, down-weights easy negatives and concentrates the gradient on hard, rare classes; we apply it in Experiments 02b, 03, and 05a, where the `O` tag dominates by a factor of about 16:1 on the training split. Discriminative or layer-wise learning-rate decay (LLRD), introduced for transfer learning by Howard and Ruder [10], applies smaller learning rates to lower transformer layers and larger rates to the task-specific head. LLRD is widely reported to stabilise DeBERTa fine-tuning and we use it in every DeBERTa-based experiment.

### E. LLMs as in-context span classifiers

Brown *et al.* [11] popularised the in-context learning paradigm in which a frozen autoregressive language model is conditioned on a task description and a small number of input–output demonstrations and produces predictions without any parameter update. Our two-stage and audit pipelines (Experiments 05a, 05b, 07–09) instantiate this recipe on `Qwen2.5-7B-Instruct` and `Gemma-3-12B-it`, run in 4-bit NF4 precision via the `bitsandbytes` integration of QLoRA [12] so that each model fits on a single 16 GB consumer GPU. The few-shot variants supply  $k=4$  gold span-label pairs per class, mined from the training split, while the zero-shot variants rely solely on the task description. As a controlled, classical-ML alternative to the LLM, we also experiment (Exp 10) with a Support Vector Machine [13] trained on encoder span embeddings; this makes it possible to test how much of the LLM’s contribution comes from world knowledge that is not already linearly accessible in the encoder’s representation.

### F. Where the field falls short

Three shortcomings recur across these works. First, the available datasets are small and English-biased; Dey *et al.* explicitly note that scaling annotation requires medical expertise and is therefore expensive [1], which is a fundamental bottleneck for any supervised approach. Second, evaluation metrics privilege exact spans, penalising systems for borderline

TABLE I  
PER-SPLIT STATISTICS OF THE SMM4H-HEARD 2026 TASK 7 CORPUS,  
AS REPORTED BY THE SHARED-TASK ORGANISERS. ENTITY COLUMNS  
COUNT *tokens*, NOT SPANS.

| Split | Posts | Social | Clinical | All entity | Tokens |
|-------|-------|--------|----------|------------|--------|
| Train | 842   | 408    | 616      | 1,024      | 17.2K  |
| Dev   | 258   | 167    | 223      | 390        | 5.2K   |
| Test  | 278   | 256    | 108      | 364        | 6.2K   |
| Total | 1,378 | 831    | 947      | 1,778      | 28.6K  |

disagreements that human experts also disagree on; relaxed F1 mitigates but does not remove this issue. Third, current encoder models exhibit weak compositional understanding of implicit references, idioms, and subreddit-specific slang — a gap that LLMs partly close at large inference cost. The four pipelines compared in this paper are designed to probe whether that cost is justified, and to map the trade-off between encoder-only structured prediction, LLM-based reasoning, and classical-ML span classifiers.

### III. METHODOLOGY

This section first describes the SMM4H-HeARD 2026 Task 7 corpus and the dataset properties that drove our design choices, and then outlines the four pipelines we built on top of it. Detailed configurations and hyper-parameters are reported in Section IV.

#### A. Dataset

1) *Source and purpose*: The dataset is the SMM4H-HeARD 2026 Task 7 corpus released by Dey *et al.* [1]. It was collected from opioid-related Reddit communities and annotated by clinical experts to support the extraction of self-reported clinical and social impacts of nonmedical opioid use. The organisers report a total of 1,378 posts and 1,778 entity tokens; per-split counts are summarised in Table I. The released CSVs that we use during training and inference contain 842 training, 258 development, and 578 test rows. Train and dev are essentially one row per post (a handful of training posts are split into multiple sentence-level rows, so 842 rows correspond to 743 unique post IDs), while the test split breaks each original post into roughly two segments so that long posts fit into the encoder’s context window. Each row contains (i) a list of *tokens*, (ii) a *ner\_tags* sequence of equal length under the BIO scheme defined in Section II, and (iii) an *ID*; the post title and source subreddit are inlined as the first few tokens of every row. The test split is unannotated and used for blind submission to the official Codabench leaderboard.

2) *Class imbalance*: Token-level statistics (Fig. 1) show that the *O* tag covers  $\sim 16,200$  training tokens against only  $\sim 1,024$  tokens for the four entity classes combined — a ratio of about 16:1. Inside the entity tokens, train and dev are mildly skewed toward *ClinicalImpacts* (616 vs 408 on train, 223 vs 167 on dev); by *span* count the skew is larger, with about three Clinical spans for every Social span on train. The test split inverts this balance: the organisers report 256 Social against only 108 Clinical entity tokens

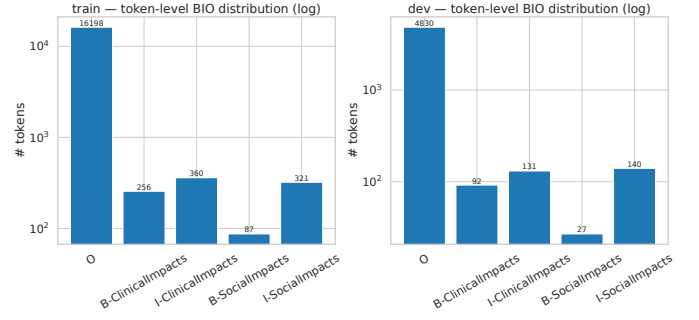


Fig. 1. Token-level BIO distribution on the train and dev splits (log scale). The *O* class dominates by a factor of  $\sim 16:1$ .

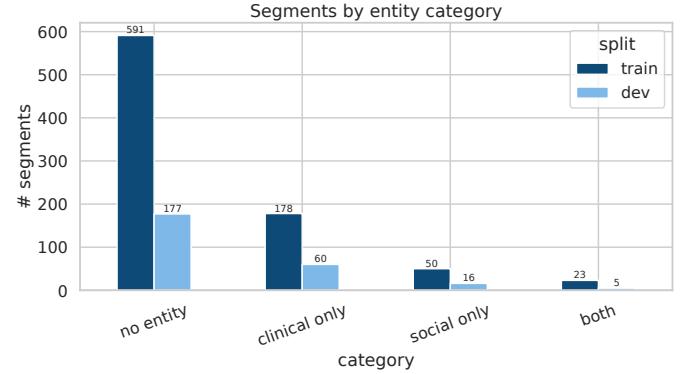


Fig. 2. Number of segments per entity category. The *no entity* class accounts for 70% of train and 68% of dev segments.

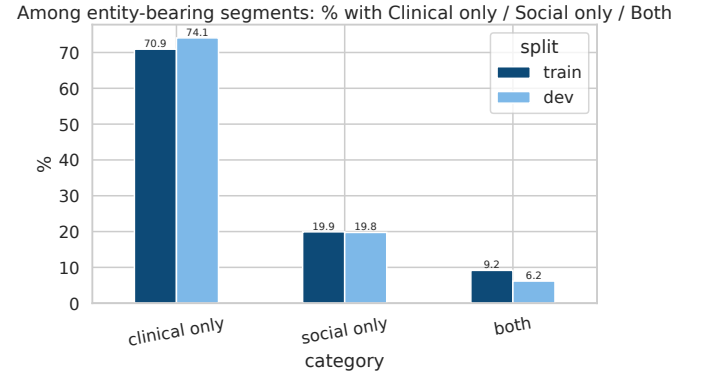


Fig. 3. Among entity-bearing segments, the share of segments containing only Clinical, only Social, or both impact types.

(Table I), so a model trained to favour the Clinical class will be penalised at evaluation time. Many segments also carry no entity at all: 591/842 training and 177/258 development segments are entirely *O* (Fig. 2). Among entity-bearing segments,  $\sim 71\%$  contain only Clinical impacts,  $\sim 20\%$  only Social, and  $\sim 9\%$  both (Fig. 3). This imbalance motivates the focal- and weighted-loss ablations of Section IV, and the use of a 3-class collapsed label set in the two-stage pipeline.

3) *Span and segment length*: Most spans are short. The histogram in Fig. 4 shows that the bulk of both *ClinicalImpacts* ( $n=256$  train, 92 dev) and *SocialImpacts* ( $n=87$  train, 27 dev) spans is between one

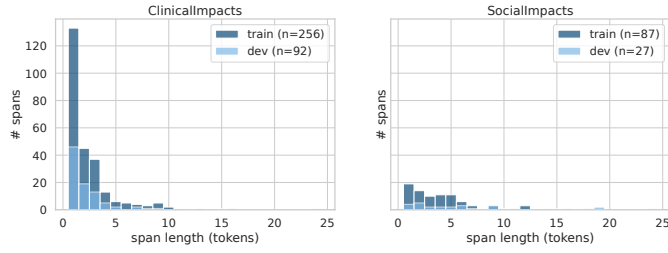


Fig. 4. Span-length distribution per entity class. Both classes are heavily concentrated in the 1–4 token range.

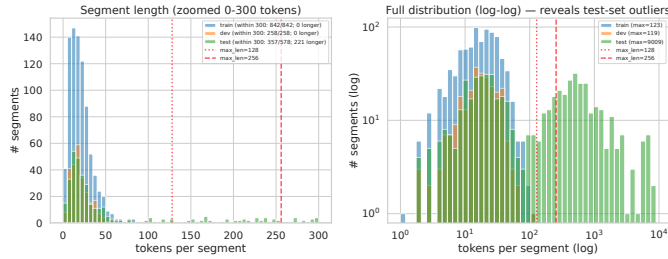


Fig. 5. Segment length in tokens (top: zoomed; bottom: log-log). Train and dev are bounded by  $\sim 120$  tokens, while 221 test segments exceed 300 tokens, with one outlier at 9,009.

and four tokens long, with a tail extending to  $\sim 25$  tokens. Train and dev segments fit comfortably within 256 tokens (max 123 and 119 respectively, Fig. 5), but the test split contains 221 segments longer than 300 tokens, including one outlier of 9,009 tokens. We therefore set the per-segment maximum length to 256 during training, and apply a sliding-window inference scheme on test segments that exceed this budget.

4) *Perspective drift between splits*: The SMM4H-HearD task targets first-person, self-reported impacts; third-person mentions (e.g. “my friend overdosed”) must remain untagged. We classified each segment by its dominant grammatical perspective using a small instruction-tuned LLM and observed a substantial mismatch between the splits (Fig. 6): train and dev each contain  $\sim 45\%$  first-person segments, while the test split is 71.1% first-person. The annotation rate is also strongly skewed by perspective: 57.7% of first-person segments contain an entity, against only 13.0% of third-person and 6.0% of other (Fig. 7). This means a system trained on the official train/dev splits sees a much larger fraction of un-annotated, off-task content than it will encounter at test time; conversely, the test split is more concentrated on the genuinely targeted perspective. We exploit neither perspective filtering nor weighting in our experiments and leave this as future work.

5) *Subreddit composition and lexical signature*: As expected, opioid-recovery communities dominate the top of the subreddit ranking (Fig. 8): suboxone, opiates, OpiatesRecovery, and quittingkratom together account for the bulk of entity-bearing segments. General-interest subreddits such as AskReddit, pics, or politics are present in the corpus but contribute almost no annotated entities, which reinforces the perspective-mismatch observation above. Word clouds of the annotated spans (Fig. 9)

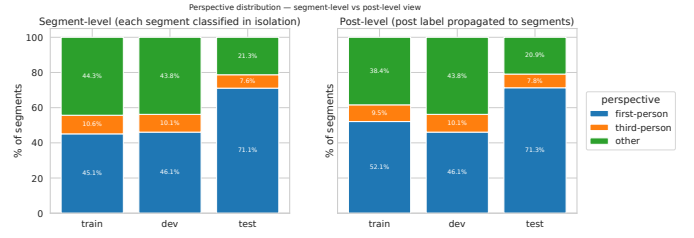


Fig. 6. Post-perspective distribution per split, evaluated both at the segment and at the post level. The test split is markedly more first-person than train or dev.

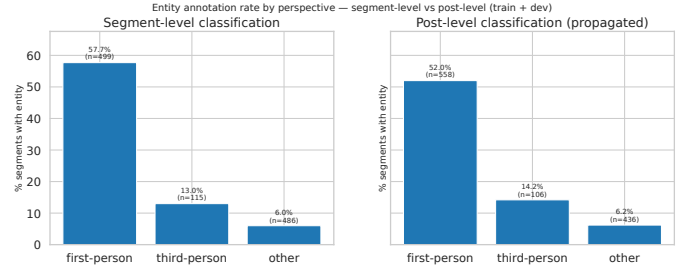


Fig. 7. Entity annotation rate by post perspective. First-person segments are  $\sim 4\times$  more likely to contain an annotated impact span than third-person ones.

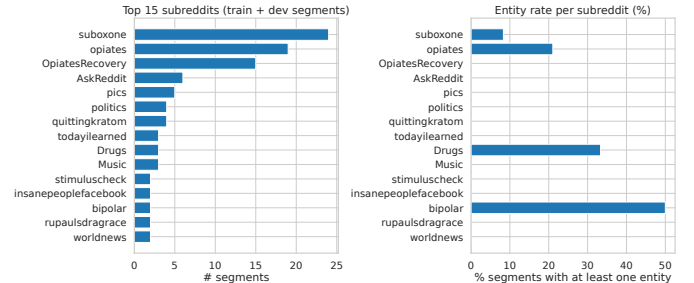


Fig. 8. Top-15 subreddits by number of train+dev segments (left) and their entity-bearing rate (right).

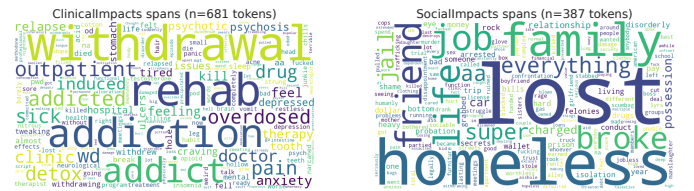


Fig. 9. Word clouds over annotated tokens, separated by entity class.

confirm the lexical distinction between the two entity types: ClinicalImpacts is dominated by tokens such as *withdrawal*, *rehab*, and *addiction*, while SocialImpacts is dominated by *lost*, *homeless*, *job*, and *family*.

## B. System design

We instantiate four pipelines, all built on top of a shared encoder-fine-tuning core. The pipelines differ only in how span boundaries and span types are recovered.

1) *Direct pipeline*: A single transformer encoder is fine-tuned on the full 5-class BIO label set and produces token-level predictions in one pass. An optional CRF head replaces

the per-token softmax with a globally normalised score over full label sequences, decoded by Viterbi at inference. This is the simplest design and serves as the encoder-only reference for the other pipelines.

2) *Two-stage pipeline*: Span detection and span typing are decoupled. A 3-class encoder is fine-tuned on a collapsed label set (O, B-Impact, I-Impact) so that all impact tokens share a single class. At inference, the encoder produces impact spans, and each span is then classified as `ClinicalImpacts`, `SocialImpacts`, or `Neither` by an external classifier. A `Neither` verdict reverts the span to O.

3) *Audit pipeline*: A 5-class encoder produces a full BIO tagging in the same way as in the direct pipeline. Every non-O predicted span is then sent to an external classifier with the encoder’s current class as a hint; the classifier may verify, correct, or reject the prediction. Unlike the two-stage pipeline, the encoder here is solely responsible for span boundaries, and the external classifier acts as a post-hoc auditor of the type assignment.

4) *External classifier*: The external classifier in the two-stage and audit pipelines is one of two backends:

- an instruction-tuned causal LLM (Qwen2.5-7B-Instruct or Gemma-3-12B-it) loaded in 4-bit NF4 precision and prompted in zero-shot or few-shot ( $k=4$  gold demonstrations per class) regimes;
- a Support Vector Machine head with PCA-128 dimensionality reduction and an RBF kernel, trained on encoder span embeddings extracted offline from the training set.

The two backends share the same call interface, which lets us swap them without changing any other part of the pipeline.

5) *Training and inference common to all pipelines*: Every encoder is fine-tuned end-to-end on the training split and selected on the development split using strict micro-F1. We use either standard cross-entropy, weighted cross-entropy, focal loss, or the CRF negative log-likelihood as training objective. Layer-wise learning-rate decay is applied to all DeBERTa-based runs. At inference, we apply the same sliding-window scheme as for training to test segments that exceed the maximum length, and we re-aggregate overlapping windows by majority vote at the token level.

#### IV. EXPERIMENTS AND OWN CONTRIBUTION

This section walks through every experiment in our ablation in the order in which one design choice motivated the next. Quantitative dev and test scores are deferred to Table III of Section V; here we focus on the rationale for each choice and on the most informative deltas. Throughout the section, “baseline” refers to Experiment 01 (Sec. IV-B).

##### A. Experimental setup common to all runs

Every encoder is fine-tuned on the official 842-row training split and selected on the 258-row development split using strict micro-F1 over the five BIO labels. We use a maximum sequence length of 256 tokens, a batch size of 16, 20 training epochs with a 10% linear warm-up, gradient clipping at 1.0, and a base learning rate of  $2 \times 10^{-5}$ . Word-level BIO labels are

propagated to sub-word tokens via the standard “first sub-token gets the B/I tag, others get -100” convention, and inference labels are extracted from the first sub-token of each word. All experiments use seed 42 unless noted otherwise. Every metric we report is computed by `segeval` on the same data, which guarantees an identical evaluation protocol across pipelines.

##### B. Baseline: encoder-only direct pipeline (Exp 01)

The baseline is a `mental/mental-roberta-base` [5] encoder fine-tuned with cross-entropy loss to predict 5-class BIO tags directly. We chose this checkpoint for two reasons: it is a RoBERTa [3] variant adapted to mental-health Reddit text, which matches our domain, and its base size ( $\sim 125$ M parameters) keeps the per-experiment training cost low enough to run a wide ablation on a single GPU. The baseline reaches 35.64% strict / 48.89% relaxed F1 on dev and 36.75% strict / 48.90% relaxed F1 on the Codabench test split. Two observations follow. First, the  $\sim 13$ -point gap between strict and relaxed scores indicates that the baseline finds entities in approximately the right places but struggles to predict their exact boundaries. Second, despite using a domain-adapted checkpoint, this baseline is roughly 12 relaxed-F1 points below the DeBERTa-large baseline reported by the shared-task organisers [1], which motivates exploring stronger backbones, structured decoders, and external classifiers.

##### C. Direct-pipeline ablation (Exp 02–04, 06)

The first experimental thread keeps the direct (single-encoder) pipeline shape and varies the loss function, the backbone, and the decoding head.

1) *Loss function (Exp 02)*: The class imbalance documented in Sec. III-A suggests that the baseline may waste capacity on the dominant O class. We re-trained the same backbone with two alternative objectives: (a) weighted cross-entropy with class weights [0.05, 1.0, 1.0, 1.0, 1.0] for {O, B-Cli, I-Cli, B-Soc, I-Soc}, and (b) focal loss [9] with  $\gamma=2$ . Weighted CE produces a small dev gain (+0.4 strict / +0.4 relaxed), but generalises much better to the test split (+4.3 strict / +6.0 relaxed over the baseline). Focal loss is essentially neutral on strict F1 and slightly negative on test strict, which suggests that explicit class weights are a more robust remedy than down-weighting “easy” negatives at this dataset scale.

2) *Stronger backbone with LLRD (Exp 03)*: We replaced `mental-roberta-base` with `microsoft/deberta-v3-base` [4], kept focal loss, and added Layer-wise Learning-Rate Decay [10] with a per-layer multiplier of 0.9 from top to bottom. LLRD is a known stabiliser for DeBERTa fine-tuning because it slows the drift of the lower (more general) layers while letting the task-specific head move quickly. The combined change boosts test scores to 42.40% strict / 55.80% relaxed, a +5.7 strict and +6.9 relaxed jump over the baseline; this is the largest single-step improvement in the entire ablation, and the relaxed score here is the highest test relaxed F1 we observe across all 16 configurations.



3) *CRF decoding (Exp 04 — main contribution)*: Inspired by Lample *et al.* [7], we add a linear-chain CRF head on top of the DeBERTa-v3 encoder, replacing the per-token cross-entropy by the CRF negative log-likelihood. The CRF rules out illegal BIO transitions (e.g. I-Cl<sub>i</sub> after B-Soc) and decodes via Viterbi at inference. The CRF gives +3.0 strict points on the test split over the LLRD baseline of Exp 03, lifting the test strict score from 42.40% to 45.38% while keeping the relaxed score nearly unchanged (55.80 → 54.20). The strict-vs-relaxed pattern is consistent with the role of the CRF: it cleans up boundaries (which strict F1 rewards) without finding new entities (which relaxed F1 rewards). Exp 04 is therefore our strongest *single-system* configuration and the encoder backbone for every downstream LLM and SVM pipeline.

4) *Multi-seed ensemble (Exp 06)*: We re-trained the Exp 04 (DeBERTa-v3-base+CRF) configuration with three random seeds (42, 123, 999) and combined the three predictions by majority vote at the token level. The ensemble reaches 44.11% strict / 55.10% relaxed F1 on test, slightly below the single-seed Exp 04. Token-level voting helps when seeds make *uncorrelated* mistakes; here all three seeds converge on the same boundary errors, so the vote has nothing to average over. We report the ensemble for completeness but use Exp 04 as our final encoder-only system.

#### D. LLM-augmented pipelines (Exp 05, 07–09)

The next experimental thread attaches an instruction-tuned LLM as an external span classifier on top of the encoder. We explore two pipeline shapes (Sec. III-B):

- *Two-stage* (Exp 05a, 07a, 08a, 09a). The encoder is re-trained with a collapsed 3-class label set ({O, B-Imp, I-Imp}) and only detects span boundaries; the LLM types each detected span as ClinicalImpacts, SocialImpacts, or Neither.
- *Audit* (Exp 05b, 07b, 08b, 09b). The 5-class DeBERTa-v3+CRF encoder of Exp 04 produces a full BIO tagging; the LLM is shown the encoder’s prediction as a hint and may verify, correct, or reject it.

For each shape we sweep two LLMs (Qwen/Qwen2.5-7B-Instruct and google/gemma-3-12b-it, both loaded in 4-bit NF4 precision via bitsandbytes [12]) and two prompting regimes (zero-shot and 4-shot, with demonstrations mined from gold training spans). Decoding is greedy, capped at 6 new tokens; the post is truncated to 2,000 characters and the candidate span to 200 characters.

The strongest LLM configuration is *audit + Gemma-3-12B + 4-shot* (Exp 09b), which reaches 42.55% strict / 56.69% relaxed F1 on dev — the highest *dev* strict score in our entire ablation. On the test split, however, Exp 09b drops to 43.20% strict / 52.30% relaxed and is decisively beaten by the encoder-only Exp 04 (45.38% / 54.20%). We attribute this inversion to the train/dev vs. test distribution shifts of Sec. III-A: the 4-shot demonstrations mined from training over-fit to the Clinical-heavy, perspective-mixed dev distribution, and on the more first-person, Social-heavy test split they no longer help.

Two further patterns emerge. First, all two-stage variants underperform their audit counterparts at relaxed F1 because the 3-class encoder finds slightly worse spans than the 5-class CRF, and the LLM cannot fully recover from upstream boundary errors. Second, 4-shot prompting consistently helps Gemma but hurts Qwen (e.g. −0.3 test strict on the audit pipeline, −0.6 on dev), suggesting that the larger Gemma model is better at extracting type signal from demonstrations while the smaller Qwen treats them more as noise.

#### E. SVM-augmented variant (Exp 10)

To isolate how much of the LLM’s contribution comes from world knowledge that is not already linearly accessible in the encoder’s representation, we replaced the LLM with a classical Support Vector Machine [13]. The classifier head is StandardScaler → PCA(128) → SVC(RBF, class\_weight=balanced), with  $C$  and  $\gamma$  chosen by 5-fold grid-searched macro-F1, and the input is the mean-pooled hidden state of the span tokens taken from the encoder of Exp 04.

The two-stage SVM (Exp 10a) achieves 43.27% strict / 43.50% relaxed on the test split: comparable strict F1 to the LLM counterparts but markedly weaker relaxed F1, because the small number of Social-class training spans (87) is not enough to robustly classify the noisy boundaries of the 3-class encoder. The audit SVM (Exp 10b) preserves most of the Exp 04 dev numbers (40.65% strict / 55.91% relaxed) and a 5-fold macro-F1 of 1.0 on gold training embeddings shows that the two classes are linearly separable in the CRF encoder’s feature space, so the SVM rarely overrides the encoder. This is a useful contrast to the LLM audits in Exp 09b: when the encoder is already strong, the LLM audit adds a measurable, if modest, dev gain (+1.3 strict over Exp 04) while the classical SVM audit adds nothing. Both, however, lose to the encoder on test.

## V. RESULTS AND DISCUSSION

Table III collects every configuration we trained or prompted, together with strict and relaxed F1 on both the development split and the official Codabench test split. Within each metric, the best score across the entire ablation is set in bold.

#### A. Headline result and overall trends

The strongest single system on the test split is the encoder-only *DeBERTa-v3-base + CRF* configuration (Exp 04), which reaches 45.38% strict / 54.20% relaxed F1 — a +8.6 strict and +5.3 relaxed gain over the *mental-roberta-base* baseline (Exp 01). The improvement is the cumulative effect of three modifications: (i) switching the backbone to DeBERTa-v3 [4] (+5.7 strict on test from Exp 02a to Exp 03 once weighted CE is replaced by focal loss with LLRD), (ii) adding a CRF head with Viterbi decoding [7], [6] (+3.0 strict on test from Exp 03 to Exp 04), and (iii) enforcing valid BIO transitions, which closes most of the strict-relaxed gap (Sec. V-C). None of the LLM-augmented or SVM-augmented variants improves on Exp 04’s test strict F1, although several stay within 1 point of it.

TABLE II  
PER-CLASS RELAXED PRECISION (P), RECALL (R), AND F1 ON THE  
DEVELOPMENT SPLIT FOR FIVE REPRESENTATIVE CONFIGURATIONS.

| Exp                      | ClinicalImpacts |       |       | SocialImpacts |       |       |
|--------------------------|-----------------|-------|-------|---------------|-------|-------|
|                          | P               | R     | F1    | P             | R     | F1    |
| 01 (m-roberta, CE)       | 0.608           | 0.493 | 0.545 | 0.529         | 0.329 | 0.406 |
| 03 (DeBERTa+LLRD)        | 0.652           | 0.462 | 0.541 | 0.664         | 0.449 | 0.536 |
| 04 (DeBERTa+CRF)         | 0.692           | 0.484 | 0.570 | 0.743         | 0.467 | 0.574 |
| 09b (Gemma audit, 4-sh.) | 0.729           | 0.471 | 0.572 | 0.743         | 0.449 | 0.560 |
| 10b (SVM audit)          | 0.671           | 0.484 | 0.563 | 0.740         | 0.443 | 0.554 |

### B. Per-class breakdown

Table II reports relaxed precision, recall, and F1 broken down by entity type for five representative configurations. The baseline shows a 14-point gap between *ClinicalImpacts* (54.5%) and *SocialImpacts* (40.6%), driven entirely by the recall side (Clinical recall 49.3% vs. Social 32.9%). This matches the training-set imbalance documented in Sec. III-A: there are  $\sim 3\times$  as many Clinical spans as Social spans, and the weakest models simply miss the Social entities. The gap closes sharply once the backbone is upgraded: by Exp 03 the two classes agree to within 0.5 F1 points, and Exp 04 actually flips the ranking (Social slightly above Clinical at 57.3% vs. 57.0%). The gap re-opens by 1–2 points in the LLM and SVM audit pipelines (Exp 09b, 10b), which suggests that the secondary classifier is biased towards Clinical and re-introduces the asymmetry the CRF had erased.

### C. Strict vs. relaxed F1

The strict score requires both the boundaries and the type to be correct, while the relaxed score only requires any token-level overlap. The gap between them therefore measures how often the system finds an entity *near* but not exactly where the annotators placed it. The baseline has a 13.3-point strict-relaxed gap on dev (35.6 vs. 48.9); this widens to 19.4 points in the LLRD configuration of Exp 03 (34.5 vs. 53.9), because the stronger encoder finds more entity *regions* but with even noisier boundaries. The CRF closes the gap back to 15.8 points on dev and to a strikingly small 8.8 points on test (45.4 vs. 54.2), confirming that the structured decoder’s main contribution is boundary cleaning rather than entity discovery. None of the LLM or SVM heads further reduces this gap, which is consistent with their span-level (rather than token-level) operating regime: by construction they cannot edit boundaries, only re-classify the spans the encoder has already proposed.

### D. Precision/recall asymmetry

A pattern shared by every system in the table is that recall, not precision, is the bottleneck. Across Exp 01–Exp 10b on dev, relaxed precision climbs from 0.58 to 0.73 as we improve the pipeline, but relaxed recall stays in the narrow 0.42–0.48 band. In other words, our systems are progressively more *conservative*: they hallucinate fewer false-positive impact spans, but the absolute number of gold spans they discover does not materially change. We attribute this to the sheer scarcity of training data — 343 Clinical and 87 Social spans

in 842 posts is too little for the model to learn the long tail of expressions for “lost my job”, “slept on the streets”, etc., so the unseen variants at test time remain invisible.

### E. Comparison to the organisers’ baseline and shortcomings

The strongest baseline reported by the shared-task organisers [1] is a DeBERTa-large model that reaches  $\sim 0.61$  relaxed F1 on the same test split. Our best system (Exp 04) is  $\sim 7$  relaxed-F1 points behind and Exp 09b is  $\sim 8$  points behind. Three factors plausibly account for the gap. First, every encoder we used is the *base* variant; given the consistent +5–+7 point gain from *mental-roberta* to *deberta-v3-base*, a further jump to *deberta-v3-large* is the most likely single source of remaining improvement. Second, our CRF and LLM audits operate on *token-level* BIO predictions; the organisers’ system uses a post-hoc span-merging heuristic that recovers contiguous entities spanning more than a single segment, which our pipeline cannot do because it processes each released CSV row independently. Third, our 4-shot demonstrations are sampled by class only; recent work on in-context learning suggests that perspective-balanced or length-balanced sampling would better match the test distribution. Addressing these points is the natural next step beyond the present study.

## VI. CONCLUSION AND FUTURE WORK

We presented a comparative study of four NER pipeline shapes on the SMM4H-HeaRD 2026 Task 7 dataset: a direct encoder, a two-stage detect-then-classify pipeline, an audit pipeline that lets a secondary classifier verify the encoder’s output, and a lightweight SVM variant of the same audit shape. Across 16 configurations, the strongest single system on the official Codabench test split is *DeBERTa-v3-base + linear-chain CRF* (Exp 04), which reaches 45.38% strict and 54.20% relaxed F1, +8.6 and +5.3 points above the *mental-roberta-base* baseline. Returning to the research questions of Sec. I, we find that: **(RQ1)** decoupling span detection from span typing does not improve test F1 over a well-decoded 5-class CRF; the best LLM- and SVM-augmented pipelines stay within 1 test strict point of Exp 04 without surpassing it. **(RQ2)** On the development split a 4-shot Gemma-3-12B audit gains +1.3 strict F1 over the encoder, but this advantage does not transfer to the test split, suggesting that span-level external classifiers are at best a marginal addition to a strong CRF backbone on this dataset. The dominant remaining error is low recall on rare social-impact expressions, driven by the small training set (87 Social spans) and the perspective-shift between train/dev and test.

The most promising directions for future work follow directly from the gap to the organisers’ DeBERTa-large baseline [1]. First, scaling the encoder to *deberta-v3-large* and adding domain-adaptive pre-training on Reddit substance-use posts would address the recall-bound regime we observed. Second, reintroducing a span-merging post-processing step that aggregates predictions across the sentence-level test rows would remove an artificial limitation of our segment-by-segment inference. Third, replacing the class-only 4-shot

TABLE III

ABLATION ACROSS ALL 16 CONFIGURATIONS. PIPELINE SHAPES: D = DIRECT, 2S = TWO-STAGE, AU = AUDIT, ENS = ENSEMBLE, 2S/AU+SVM = SVM-AUGMENTED TWO-STAGE/AUDIT. CRF AND LLRD COLUMNS ARE Y/N. "EXTERNAL" GIVES THE SPAN CLASSIFIER USED IN TWO-STAGE, AUDIT, AND SVM PIPELINES. BEST PER METRIC IN BOLD.

| Exp | Pipe   | Backbone                   | Loss    | CRF | LLRD | External               | Dev S         | Dev R         | Test S        | Test R        |
|-----|--------|----------------------------|---------|-----|------|------------------------|---------------|---------------|---------------|---------------|
| 01  | D      | mental-roberta-base        | CE      | N   | N    | —                      | 0.3564        | 0.4889        | 0.3675        | 0.4890        |
| 02a | D      | mental-roberta-base        | W-CE    | N   | N    | —                      | 0.3603        | 0.4928        | 0.4096        | 0.5490        |
| 02b | D      | mental-roberta-base        | Focal   | N   | N    | —                      | 0.3387        | 0.4924        | 0.3362        | 0.5070        |
| 03  | D      | deberta-v3-base            | Focal   | N   | Y    | —                      | 0.3446        | 0.5386        | 0.4240        | <b>0.5580</b> |
| 04  | D      | deberta-v3-base            | CRF NLL | Y   | Y    | —                      | 0.4130        | <b>0.5714</b> | <b>0.4538</b> | 0.5420        |
| 06  | Ens    | deberta-v3-base $\times 3$ | CRF NLL | Y   | Y    | —                      | 0.4098        | 0.5649        | 0.4411        | 0.5510        |
| 05a | 2S     | deberta-v3-base (3-cls)    | Focal   | N   | Y    | Qwen2.5-7B, zero-shot  | 0.3941        | 0.4230        | 0.4206        | 0.4490        |
| 05b | Au     | deberta-v3-base            | CRF NLL | Y   | Y    | Qwen2.5-7B, zero-shot  | 0.3860        | 0.5407        | 0.4426        | 0.5490        |
| 07a | 2S     | deberta-v3-base (3-cls)    | Focal   | N   | Y    | Qwen2.5-7B, 4-shot     | 0.3814        | 0.4309        | 0.4340        | 0.4830        |
| 07b | Au     | deberta-v3-base            | CRF NLL | Y   | Y    | Qwen2.5-7B, 4-shot     | 0.3797        | 0.5369        | 0.4400        | 0.5520        |
| 08a | 2S     | deberta-v3-base (3-cls)    | Focal   | N   | Y    | Gemma-3-12B, zero-shot | 0.3824        | 0.4207        | 0.3946        | 0.4180        |
| 08b | Au     | deberta-v3-base            | CRF NLL | Y   | Y    | Gemma-3-12B, zero-shot | 0.3877        | 0.5502        | 0.4000        | 0.4960        |
| 09a | 2S     | deberta-v3-base (3-cls)    | Focal   | N   | Y    | Gemma-3-12B, 4-shot    | 0.4076        | 0.4376        | 0.4279        | 0.4440        |
| 09b | Au     | deberta-v3-base            | CRF NLL | Y   | Y    | Gemma-3-12B, 4-shot    | <b>0.4255</b> | 0.5669        | 0.4320        | 0.5230        |
| 10a | 2S+SVM | deberta-v3-base (3-cls)    | Focal   | N   | Y    | SVM (RBF, PCA-128)     | 0.3571        | 0.3674        | 0.4327        | 0.4350        |
| 10b | Au+SVM | deberta-v3-base            | CRF NLL | Y   | Y    | SVM (RBF, PCA-128)     | 0.4065        | 0.5591        | —             | —             |

demonstration sampler with a perspective and length-aware retriever would mitigate the dev/test prompt mismatch we identified.

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